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On Curvature and Orthogonal Surfaces (continued).

[Continuation; book pages 304-315.]

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or conversely

$$\begin{aligned}\xi' &= (1 + a_1)\alpha\xi + d_{12}\alpha\eta + d_{13}\alpha\zeta, \\ \eta' &= d_{21}\beta\xi + (1 + d_{22})\beta\eta + d_{23}\beta\zeta, \\ \zeta' &= d_{31}\gamma\xi + d_{32}\gamma\eta + (1 + d_{33})\gamma\zeta,\end{aligned}$$

say $\xi', \eta', \zeta' = \xi + \Delta\xi, \eta + \Delta\eta, \zeta + \Delta\zeta$; hence

$$\begin{aligned}X'\xi' + Y'\eta' + Z'\zeta' &= (X' - V d_1x)(\xi + \Delta\xi) + \&c. \\ &= X\xi + Y\eta + Z\zeta \\ &\quad + X\Delta\xi + Y\Delta\eta + Z\Delta\zeta \\ &\quad - V(d_1x\xi + \omega d_2y\eta + d_3z\zeta),\end{aligned}$$

where second line is

$$\begin{aligned}&(Xa + Y\beta + Z\gamma)(d_1x\xi + \omega d_2y\eta + d_3z\zeta) \\ &+ \mu\{X d_1y\eta + Y d_2y\xi + \&c.\} + (X\Delta_1\xi + Y\Delta_2\eta + Z\Delta_3\zeta) + \omega(X\Delta_2\xi + Y\Delta_3\eta + Z\Delta_1\zeta).\end{aligned}$$

But

$$\begin{aligned}X\Delta_1\alpha + Y\Delta_2\beta + Z\Delta_3\gamma &= \frac{\partial\Sigma}{\partial\xi} - V d_1X = 0, \\ X\Delta_2\alpha + Y\Delta_3\beta + Z\Delta_1\gamma &= 0, \\ X\Delta_3\alpha + Y\Delta_1\beta + Z\Delta_2\gamma &= 0,\end{aligned}$$

or second line is $= V(d_1x\xi + \omega d_2y\eta + d_3z\zeta)$; we have therefore

$$X'\xi' + Y'\eta' + Z'\zeta' = X\xi + Y\eta + Z\zeta;$$

We require

$$(A', B', C', F', G', H'\sim\xi', \eta', \zeta')^2;$$

viz. to the first order Δ , this is

$$\begin{aligned}&= (A, \dots \sim\xi, \eta, \zeta)^2 \\ &+ 2(\Delta \dots \sim\xi, \eta, \zeta)^2 + \&c. \&c.,\end{aligned}$$

28. Here second line is

$$2\{A\xi + B\eta + G\xi \Delta\xi + (H\xi + B\eta + F\xi)\Delta\eta + (G\xi + F\eta + C\xi)\Delta\zeta\}$$

but

$$H\xi + B\eta + G\zeta = B\eta - VH\xi$$

$$\begin{vmatrix} \alpha, & \beta, & \gamma \\ X, & Y, & Z \\ \xi, & \eta, & \zeta \end{vmatrix}$$

$$H\xi + B\eta + F\zeta = B\eta - VH\xi$$

$$\begin{vmatrix} \alpha, & \beta, & \gamma \\ X, & Y, & Z \\ \xi, & \eta, & \zeta \end{vmatrix}$$

$$G\xi + F\eta + C\zeta = VH\xi + Xh\eta$$

$$\begin{vmatrix} \alpha, & \beta, & \gamma \\ X, & Y, & Z \\ \xi, & \eta, & \zeta \end{vmatrix}$$

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whence term in $\{ \}$ is

$$\begin{vmatrix} d\xi \Delta\xi, & d\xi \Delta\eta \\ d\eta \Delta\xi, & d\eta \Delta\eta \\ X, & Y, & Z \end{vmatrix} + \text{a sum of similar terms,}$$

which might be written

$$\begin{vmatrix} d\xi, & d\eta, & d\zeta \\ \Delta\xi, & \Delta\eta, & \Delta\zeta \\ X, & Y, & Z \end{vmatrix} = \begin{vmatrix} \xi, & \eta, & \zeta \\ X, & Y, & Z \end{vmatrix}$$

but it is perhaps more convenient to retain the second term in its original form.

29. As regards the first line, we have

$$\begin{aligned} A &= BC' - F^2V \\ &= V(b + \Delta b)(F + V d_1h) - \{(g + V d_2g)(F - V d_3g)\} \\ &= A + V(\Delta_1b d_1h - 2 d_2g d_3g + 2gh\xi); \end{aligned}$$

with similar expressions for the other coefficients. Attending only to the terms of the first order, we then obtain

$$\begin{aligned} A &= A + 2(B \Delta_1b - Vgh) - 2V(d_1h\xi - gh\xi), \\ B &= B + 2(C \Delta_1c - Vfg) - 2V(d_2h\xi - gh\xi), \\ C &= C + 2(A \Delta_1a - Vfh) - 2V(d_3h\xi - gh\xi), \\ F &= F + V d_1\alpha - X d_2\alpha - Y(d_2c - \Delta\eta) - V(\xi d_1h\eta - gh\xi - gh\eta) + \xi h\eta, \\ G &= G + V d_1\beta - X d_1\alpha - V(\Delta b - \Delta\eta) - V(d_2h\xi - gh\xi - gh\eta) + gh\eta, \\ H &= H + X d_2\alpha - X d_1\beta - V(\Delta c - \Delta\xi) - V(d_3h\xi - fgh\xi - gh\eta) + gh\eta; \end{aligned}$$

my these are $A' = A + bA$, &c., where b is a functional symbol; we then have

$$(A' \dots \xi', \eta', \zeta')^2 = (A \dots \xi, \eta, \zeta)^2 + (bA \dots \xi, \eta, \zeta)^2 + 2(A \dots \xi, \eta, \zeta)(\Delta\xi, \Delta\eta, \Delta\zeta);$$

which, for shortness, I represent by

$$(A \dots \xi, \eta, \zeta)^2 + (bA \dots \xi, \eta, \zeta)^2,$$

and I proceed to complete the calculation of the coefficients A' , B' , &c.

30. We have

$$\begin{aligned} A' &= bA + \text{coeff. } b'^2 \text{ in} \\ &2\{B \Delta b + G\xi \Delta\eta + (H\xi + B\eta + F\xi)\Delta_1h + (G\eta + F\eta + C\xi)\Delta_3h\} \\ &= bA + 2\{aA\eta \Delta_1\alpha + 2gh\eta\eta\} \end{aligned}$$

that is,

$$\begin{aligned} A' &= bA + \frac{2}{V}(AX + HY + GZ) d_1\alpha \\ &+ 2\eta(d_2b \Delta_2h + h\xi \Delta_3h) \end{aligned}$$

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where coeff. $b\eta$ is

$$\begin{aligned} &= \frac{A\eta + Hb + G\eta}{V} - \frac{(AX + HY + GZ)\eta\beta\eta}{V} \\ &= \frac{1}{V} \left\{ (a\alpha - p\beta) - p^2\alpha + (\gamma + b)\beta + (gg + Gh) d_2 \right\}. \end{aligned}$$

31. And similarly,

$$\begin{aligned} F' &= bF + V(\Delta a - \Delta\eta) + V d_2g + V d_3g + V(d_1\alpha + b\eta\xi h + gh\xi + h\eta\xi\xi h) d\eta_c d\eta_c\eta + d_3h d_c d_c\eta \\ &\quad + p(dXYh + ZV F d_1h) d\eta_c + (\xi h + gh) d_2 d\eta_c + (Xb + HY + FZ) d_2h d_c \\ &\quad - 6F d_1h - ph\xi - h(d_c\eta\eta\eta) d\eta, \end{aligned}$$

which gives the following formulæ; viz. on these lines

$$\begin{aligned} F' &= bF + V(\Delta_2 YZ - EXY) + \frac{1}{p} \left\{ p(XHX + GLY) d_c\eta_c - V d d_2 \right\} \\ &\quad - 2(VF d_1\alpha + V d d_c + V d d_c\eta + \&c) \\ &\quad + 2V(d_c\eta + V d d_c\eta) d_c\eta \\ &\quad - 2V(db\eta - p\xi) d_c\eta, \end{aligned}$$

which to denote in the foregoing equation, together with the following terms:

$$= \frac{1}{p}(\alpha A + V d \alpha) d_c\eta + 2(db\eta - p\xi) d_c\eta,$$

which destroy certain of the foregoing ones; viz. we have

$$A' = (b + \frac{2FHX}{V}) d_c\eta + 2(VH - VXY) d_c\eta.$$

32. Similarly, the value of bF is

$$\begin{aligned} bF &= V \left\{ -\frac{p}{V}(\alpha + h) + d_c\eta XY \right\} = \frac{1}{p} \left\{ V(YHX + LXY) d_c\eta - V d d\xi \right. \\ &\quad \left. - 2VY \left(\frac{1}{p} d_cp - p\beta \right) \xi d\eta + V d d\xi \right. \\ &\quad \left. - 2VY(d_c\eta - p\beta) - 2VY(d\xi - d_c\xi) d\eta + V d d\xi \right. \\ &\quad \left. + p dV d\xi \right. \\ &\quad \left. - p\xi d\eta d_c + p\xi d\eta, \right. \end{aligned}$$

which is

$$\begin{aligned} &= \frac{1}{p}(\alpha b\eta + Vh) \left[\frac{p}{V}(YHX - EXY) + \frac{FVHX}{V}(VHX - XHY) + \frac{p}{V}d\xi d\eta \right] \\ &\quad + \left\{ -\frac{1}{p}(B\xi + EXY + Vh - h) \right\} \eta d_c\eta \\ &\quad + \left\{ \frac{pX}{V} - \frac{2HY}{V} - Vh \right\} d\eta d_c\eta \\ &\quad + (VY d\xi + VX d\eta + VXH\eta - VXY)p \end{aligned}$$

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The value of θA is

$$\begin{aligned}\theta A &= 2E \left\{ -\frac{p}{V} (b\alpha + h) + \frac{1}{p} d_c EXY \right\} = \frac{1}{V} (2Vh\eta + LX\eta d\xi - Vd d\xi) \\ &\quad - 2VY \left\{ -\frac{p}{V} (b\eta + h) + \frac{1}{p} d_c \eta EXY \right\} = \frac{1}{V} (GBX + LX\eta d\xi - Vd d\xi) \\ &\quad - 2V(bh\xi - p\eta\xi),\end{aligned}$$

which is

$$\begin{aligned}&= \frac{p}{V} (2b\alpha + b\eta) + \frac{p}{V} (XY - YX) - \frac{pX}{V} (XY - EXY) \\ &\quad - \frac{2b\xi}{V} (BX - Vh) - 2V(b\eta d\xi - p\xi d\eta).\end{aligned}$$

Hence the value of A' is equal to the last-mentioned expression, together with the following terms:

$$= \frac{p}{V} (\alpha A + b\eta) + \frac{2XZ}{V} \left[(b\xi) d_c \eta + 2F(d\xi - YH) d\xi - V(d\xi - YX) d\eta \right]$$

which destroy certain of the foregoing ones; viz. we have here

$$A' = (\theta + \frac{2FHX}{V}) \xi d\xi + 2(VH - VXY) d\xi.$$

And similarly, the value of δF is

$$\begin{aligned}\delta F &= V \left\{ -\frac{p}{V} (b\alpha + h) + \frac{1}{p} d_c LXY \right\} d\xi (LH + EXY) d\xi = Vd d\xi \\ &\quad - 2 \left\{ \frac{p}{V} (LH + LH) + \frac{pLH}{V} \right\} d\xi d\eta + V(LEH - V) d\xi \\ &\quad - 2 \left\{ \frac{pXH}{V} - \frac{2HY}{V} - Vh \right\} d\xi d\eta - \frac{pH}{V} (Vh - LH) - Vh LH d\xi \\ &\quad - 2 \left\{ \frac{p}{V} - 2Vh \right\} d\xi d\eta - Vd h d\xi + VXY d\xi - VXY d\xi\end{aligned}$$

which is

$$\begin{aligned}&= \frac{p}{V} (2b\alpha - b\eta) + \frac{p}{V} (YHX - LXY) + \frac{pHY}{V} (YHX - XHY) \\ &\quad + \left\{ -\frac{1}{p} (\alpha LH + LXY) d\xi - Vh \right\} d\xi \\ &\quad + \left\{ \frac{pL}{V} (LH + LH) + 2Vh \right\} d\xi \\ &\quad + \left\{ \frac{pLH}{V} - LH \right\} d\eta \\ &\quad + \left\{ -VV d_c \eta + VBX d_c \eta + VX\eta d_c \eta - VH d_c \eta \right\} d\eta\end{aligned}$$

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Hence F' is equal to the foregoing expression, together with the following terms:

$$\begin{aligned} & + \frac{p}{V} \left(\beta A + \frac{2FHY}{V} \right) (XLE - EXY) - \frac{pHY}{V} (YXX - XXX) \\ & + \frac{p}{V} (\beta A - 2EXY) + \frac{p}{V} (XLE - EXY) \\ & + \frac{pHL}{V} (LH + GLY) d\xi + \frac{pL}{V} (LXE + LXY) d\eta, \end{aligned}$$

which destroy certain of the foregoing; viz. we then have

$$\begin{aligned} F' &= bF + V(YE - EXY) + \frac{pHXY}{V} d\xi d\eta + \frac{p}{V} d\xi + V d\xi d\eta + \frac{HLXY}{V} d\xi \\ &+ VY d_c\eta d\eta + VHYH d_c\eta + VHYXH d_c\eta. \end{aligned}$$

34. We have here

$$A' = 2 \left(Yh - \frac{VHX}{V} \right) d\xi + 2 \left(Yh - \frac{VHX}{V} \right) d\xi + 2 \left(Yh - \frac{VHY}{V} \right) d\xi + 2V(Vh - XYh) d\xi + V(VY - VZ) d\xi$$

$$B' = -2 \left(Y - \frac{VHY}{V} \right) d\xi + 2 \left(Yh - \frac{VHY}{V} \right) d\xi + 2V(Vh - XYh) d\xi - VYh d\xi + EXYh d\xi$$

$$C' = -2 \left(Y - \frac{VHX}{V} \right) d\xi + 2 \left(Yh - \frac{VHY}{V} \right) d\xi + 2V(Yh - XYh) d\xi$$

$$F' = \left\{ V(h-1) + \frac{p}{V} (YE - LXY) d\xi + \left(Yh - \frac{VHY}{V} \right) d\xi + V(Vhh - LXY) d\xi d\eta \right\} - VY_a d\xi - 2bH d\xi - 2V(bX) d\xi$$

$$G' = \left(Yh - \frac{pXY}{V} \right) d\xi + \left\{ V(h-1) - \frac{1}{p} (YXH - EXY) d\xi \right\} + V(Vhh - VYh) d\xi + VY_a d\xi - 2bV d\xi - 2V(VX) d\xi$$

$$H' = \left(Yh - \frac{VHX}{V} \right) d\xi - \left(Yh - \frac{VYH}{V} \right) d\xi + V(h-1) - \frac{1}{V} (LXH - VXH) d\xi + V(VXVh - VVX) d\xi + V d\xi$$

35. It will be recollected that we have $X'\xi' + Y'\eta' + Z'\zeta' = X\xi + Y\eta + Z\zeta$; by what precedes it appears that for the given surface the principal tangents are determined by the equations

$$(A \dots \xi, \eta, \zeta)^2 = 0,$$

$$X\xi + Y\eta + Z\zeta = 0,$$

and that the lines which (in the tangent plane of the given surface) correspond to the principal tangents of the corresponding point of the vicinal surface are determined by the equations

$$(A \dots \xi, \eta, \zeta)^2 + (\Delta \xi, \eta, \zeta)^2 = 0,$$

$$X\xi + Y\eta + Z\zeta = 0.$$

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Condition that the two surfaces may belong to an Orthogonal System.

Art. Nos. 36 to 44.

36. The condition in order that the two surfaces may belong to an orthogonal system is that the principal tangents shall correspond, or, what is the same thing, the lines which (in the tangent plane of the given surface) correspond to the principal tangents of the vicinal surface be the principal tangents of the given surface. When this is the case, the plane and now $X\xi + Y\eta + Z\zeta = 0$, $(\Delta \dots \xi)^2 = 0$, $(\xi'\eta'\zeta')$ intersect in the principal tangent, and this is therefore the required condition.

The plane $X\xi + Y\eta + Z\zeta = 0$ meets the cone $(A \dots \xi)^2 = 0$ in the principal tangents, that is, in pair of lines harmonically related to the circular lines and also to the chief tangents. Forming the two coefficients (A') , (B') , (C') , (F') , (G') , (H') of these terms in the same way as (A) , &c. are formed from A , &c., they must therefore as it is the case for the chief tangents satisfy

$$(A')\xi^2 + (B')\eta^2 + (C')\zeta^2 = 0,$$

$$(F')\xi, \quad (G')\eta\xi, \quad \dots = 0,$$

or, what is the same thing,

$$(A' \dots)\xi^2 = 0,$$

$$(A' \dots)(\dots)\xi^2 = 0.$$

The former of these, as about to be shown, is an identical identity; we have therefore the second of these, say $A' \dots$, &c. = 0 as the required condition.

37. We have

$$(A') + (B') + (C') = A + B + C + V^2 - (A \dots \xi X, Y, Z)^2,$$

$$B^2 + F^2 + C = (BC - F^2V)X^2 + \dots = V(XHYE\xi - XXHY)^2 - V(XX - HX)^2,$$

Forming next the expression of $A'X + B'Y + C'Z$, &c., and for convenience using down separately the terms which involve the second differentials of X , Y , Z , we have

$$\begin{aligned} A'X + B'Y + C'Z &= (AX + HY + GZ) d_1\alpha \\ &+ d_2\alpha V(YXY\alpha) - V(XBXY - VXYh) d\xi - V(XXY - YXYh - VXYVh) d\eta d\xi \\ &= V(XXYZH - VXYh)\xi \\ &- 2V(VHXXH - VXYVh - VX)\xi d\xi d\eta + (XXY - VXYh) d\eta d\xi \end{aligned}$$

where AV stands for $\frac{1}{V}(LX\xi + Vh\xi + VH)$, and so on, and where these expressions contain also the following terms respectively:

$$\begin{aligned} &-VZ d_c \bullet + (X d_1\alpha - p\xi)X d_c\eta + X d_c\eta d_c - VZ d_c\xi d_c\eta\eta \\ &+ ZE d_c\eta - p\xi d_c\eta + (\alpha - p\xi) d_c\eta + VX\xi d\xi d\eta d\xi \\ &+ XX d_c\eta + ZX d_c\eta + p\xi d_c\eta - VXY d_c\eta - V(dZ + Vp) d\xi \end{aligned}$$

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Multiplying by X, Y, Z and adding, the terms which contain the second differential coefficients disappear, and we obtain

$$(A' \dots X, Y, Z)^2 = V^2 \{BXY - YEZ\} d_c \alpha + (XKH - EZX) d_c \beta + (YKH - YHY) d_c \gamma \}$$

so that, attending to the above value of $A^2 + B^2 + C^2$, we have the required equation

$$(A' \dots) d_c X, Y, Z)^2 = 0.$$

38. Proceeding now to form the value of $(A' \dots)(\dots)$, that is,

$$A'(\xi\eta + B'\eta\beta + C'\zeta\beta + 2F'\eta\zeta + 2G'\zeta\xi + 2H'\xi\eta),$$

as regards those containing the first differential coefficients of ρ , namely of themselves; as regards those containing the second differential coefficients, forming the auxiliary equations

$$\begin{aligned} (A) &= 2(b)A - 2(g)V, \\ (B) &= 2(f)X - 2(h)V, \\ (C) &= 2(g)V - 2(f)Z, \\ (F) &= -(h)Y - (g)Z - (l)b + (c)Y, \\ (G) &= -(f)X - (g)Y + (l)b + (c)X, \\ (H) &= -(g)X - (f)Y + (h)b + (g)X, \end{aligned}$$

we find without difficulty that the terms in question (being, in fact, the complete value of the expression) are

$$= V(\xi A \dots d_a, d_b, d_c)\rho.$$

39. As regards the terms involving the first differential coefficients, observe that the whole coefficient of $b\rho$ is

$$\begin{aligned} &= 2(\xi) \left(Yh - \frac{VHX}{V} \right) \\ &\quad + 2(\xi) \left(Vg - \frac{VHX}{V} \right) \\ &\quad + 2(\xi) \left(V(h - h) - \frac{p}{V} (YXY - EXY) \right) \\ &\quad + 2(\xi) \left(Yh - \frac{VHX}{V} \right) \\ &\quad + 2(\xi) \left(Yg - \frac{VHX}{V} \right) \end{aligned}$$

which is

$$\begin{aligned} &= 2V \left\{ (g)(\Delta + (g) + (l)b + (h)b + (l)f) \right\} b \\ &\quad + \frac{2}{V} (\Delta(\xi b) + (l)LX + (g)(LXH + (f)LX + (h)LX) \} b. \end{aligned}$$

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40. The reduction depends on the following auxiliary formulæ:

$$\begin{aligned} a(g\eta + h)\Delta + p(g\eta - YH - XLX)(\eta b) + (l)(\xi) + (l)\xi + (l)(\xi) &= -XEX, \\ l + g(\xi)h + (l)(\xi) &= -YEX, \\ a + b + (l) + \dots &= -ZLX, \\ b(\xi)k - q + V \dots &= -XEX, \\ y + g(\xi)h + (l)(\xi) &= -EXY, \\ g(\xi)k - q + V \dots &= -YEX, \end{aligned}$$

where, for shortness, I have written LX , LY , LZ to stand for $aX + gY + bZ$, $\xi X + bY + \xi Z$, $gX + \xi Y + gZ$ respectively, and YXY for $bX + YdZ$, $dZX + YdZ$ for $dX + dZ$, etc. etc. for dX , dY , dZ .

From these we immediately have

$$\begin{aligned} (\xi)LX + (l)LY + (h)LZ &= V(LX\eta - YEX), \\ (h)LX + (l)LY + (l)LZ &= V(LXdY - YEY), \\ (g)LX + (l)LY + (l)LZ &= V(LXdZ - YEZ), \end{aligned}$$

Hence, in the coefficient of $b\rho$, the first term is

$$= 2V(-YEX + YEX + YEX),$$

and the second line is

$$= \frac{1}{V}\{(VX(LX + \xi)LX) + 2V(VLX - VEX)(LX)\}.$$

so that the sum, or whole coefficient of $b\rho$, is $= 0$. Similarly, the coefficients of $b\rho$ and $\partial\rho$ are each $= 0$.

41. We have thus arrived at the equation

$$(A \dots \check{d}_a, d_b, d_c)\rho = 0$$

as the condition to be satisfied by the normal distance ρ in order that the given surface and the vicinal surface may belong to an orthogonal system; viz. this is a partial differential equation of the second order, its coefficients being given functions of $X, Y, Z, a, b, c, f, g, h$, the first and second differential coefficients of r (where $r = r(x, y, z)$ is the equation of the given surface).

The equation, it is clear, may also be written in the two forms

$$(A \dots \check{d}_a Y_c Z_c, Yd_a, Zd_b, Xd_a \check{\rho})\rho = 0,$$

and

$$\left| \begin{array}{ccc} F^2 & G^2 & dXd_b \\ e\Gamma + bGd_c + phX + eGd_c fX & pd_1 + bhX & \frac{F^2}{V} \end{array} \right| \rho = 0,$$

if, for shortness, P, Q, R are written to denote $dZ_c - Yd_c d_c, Xd_a d_c, Yd_c d_c, Xa d_c$, respectively, it being understood that on each of these forms the d_a, d_b, d_c operate on the ρ only.

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Condition that a family of surfaces may belong to an Orthogonal System.

Art. Nos. 42 to 43.

42. We pass at once to the condition in order that the family of surfaces

$$r = r(x, y, z) = 0$$

may belong to an orthogonal system, viz. when the vicinal surface belongs to the family, we have ρ proportional to $\frac{1}{V} \left(= \frac{1}{\sqrt{X^2 + Y^2 + Z^2}} \right)$, and the condition is

$$(A \dots \sim) d_a, d_b, d_c \sim \frac{1}{V} = 0,$$

where r is a function of (x, y, z) , the first and second differential coefficients of which are $X, Y, Z, a, b, c, f, g, h$, and the equation is thus a partial differential equation of the third order satisfied by r . The form is by no means an inconvenient one, but it admits of further reduction.

43. We have $d_a \frac{1}{V}, d_b \frac{1}{V}, d_c \frac{1}{V}$ equal to $-\frac{1}{V^2} aX, -\frac{1}{V^2} LY, -\frac{1}{V^2} LZ$ respectively, and thence

$$d_a \frac{1}{V} = -\frac{1}{V^2} (aX + b\xi + g\xi d_a \xi) + \frac{3X}{V^4} (LX),$$

$$d_c d_a \frac{1}{V} = -\frac{1}{V^2} (g\xi + h + f d_a \xi) d_c + \frac{3X}{V^4} LXY,$$

or, as these may be written,

$$d_a d_c \frac{1}{V} = -\frac{1}{V^2} (\xi a + b\xi + b\eta) d_c + \frac{3X}{V^4} (LX),$$

$$d_c d_c \frac{1}{V} = -\frac{1}{V^2} (g\xi + g\eta + g d_c) d_c \eta + \frac{3X}{V^4} LXY,$$

with the like values for $d_c \frac{1}{V}$, &c. Substituting, the equation contains a term multiplied by a , viz. this is

$$= -\frac{1}{V^2} (A \dots) d_c,$$

which vanishes; and a term multiplied by a , viz. this is

$$= \frac{6}{V^4} (LX + (\beta) + (\gamma)) V,$$

which also vanishes. Writing down the remaining terms, and multiplying the whole by $-V$, the equation becomes

$$(A \dots \sim)(\xi), \dots \sim(\xi), \dots \sim \dots = -\frac{1}{V^2} (A \dots \sim LX, LY, LXY \sim = 0.$$

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44. The last term admits of reduction; from the equations

$$(A) = AV^2 + 2YXV - 2XYE, \text{ \&c., we find}$$

$$(A) LX + (H) LY + (G) LZ = (AbY + BLY + GLZ) + VPF(XBLY - YLZ),$$

$$(H) LX + (B) LY + (F) LZ = (HXY + BLY + FLZ) + VPF(YGLZ - ZLX),$$

$$(G) LX + (F) LY + (C) LZ = (GLX + FLY + CLZ) + VPF(YLZ - XLY),$$

and hence

$$(A \dots \sim LX, LY, LXY) = V^2 (A \dots \sim LX, LY, LZ),$$

wherefore the equation becomes

$$(A \dots \sim)(\xi), \dots \sim + 2 (A \dots \sim LX, LY, LZ) = 0.$$

45. It will be shown that we have identically

$$(A \dots \sim)(\xi), \dots \sim = -(A \dots \sim \xi X, \xi Y, \xi Z) \begin{vmatrix} LX, & XY, & LZ \\ X, & Y, & Z \\ LX, & LY, & LZ \end{vmatrix}.$$

The partial differential equation thus assumes the form

$$(A \dots \sim \xi) \dots = 0,$$

where Ω may be expressed indifferently in the three forms,

$$\begin{aligned} &= 4 (A \dots \sim \xi) \dots, \\ &= 4 (A \dots \sim LX, LY, LZ), \\ &= -4 \begin{vmatrix} LX, & LY, & LZ \\ X, & Y, & Z \\ LX, & LY, & LZ \end{vmatrix}. \end{aligned}$$

46. Taking the first of these, the partial differential equation is

$$(A \dots \sim \xi) \dots = 2 (A \dots) \sim \xi = 0,$$

or, written at full length, it is

$$\begin{aligned} &(A) ba + (B) bb + (C) bc + 2(F) bf + 2(G) bg + 2(H) bh \\ &- 2(A, \dots \sim LX, LY, LZ) + (A, \dots \sim ba, bb, bc, bf, bg, bh) = 0, \end{aligned}$$

where the coefficients are given functions of $X, Y, Z, a, b, c, f, g, h$, the first and second differential coefficients of v ; and it is written to denote $Xd_a + Yd_b + Zd_c$, that is, $bX = Xd_a X + Yd_b X + Zd_c X$, and similarly for the others bY, bZ .

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47. It remains to prove the above-mentioned identity.

To reduce the term $(A \dots \xi X, LY, LZ)^2$, we have

$$\begin{aligned} & A\xi X + HLY + GLZ \\ &= A(aX + bY + gZ) + H(bX + bY + fZ) + G(gX + fY + cZ) \\ &= X[a(LY^2 - F^2) + b(HY - GZ) + g(GE - YH)] \\ &\quad + Y[a(LY - YH) - b(LX - YH) + g(GE - YH)] \\ &\quad + Z[a(LX - GE) + b(LX - HX) - g(LX - YH)] \\ &= a(LV^2 - XYE) - b(LV^2 + ZEX) - g(LV^2 + YHY); \end{aligned}$$

that is,

$$A\xi X + LV^2 + GLY = XV^2 - a(YEZ + ZLY) + b(ZLZ - XLX),$$

and similarly

$$HLX + BLY + FLZ = (YEZ - LXY) + V(a(XEX - XEY) + h(LX - LX)),$$

$$GLX + FLY + CLZ = -a(LX - LX) + V(a(XEY - YH) - g(LX - LX)),$$

whence

$$\begin{aligned} (A, B, C, F, G, H \dots LX, LY, LZ)^2 &= V^2(LX + YLY + ZLZ) \\ &\quad - 2(a(LY - LXY)(LXY - LXZ) + LXYh(LY - LXZ)), \end{aligned}$$

48. Now, from the equations $AX + HY + GZ = -V\xi$, &c., we have for the value of even the foregoing determinant

$$\begin{aligned} & \{2d\xi_c d\eta - 2bX d\eta bX + 2(b\eta)b(bX + aY d\xi)\}b \\ &= (BP + CY^2 - 2FYZ)X^2 \\ &\quad + (CX^2 + AZ^2 - 2GZX)Y^2 \\ &\quad + (AY^2 + BX^2 - 2HXY)Z^2 \\ &\quad + 2(-AYZ + FX^2 + GXY + HXZ)YZ \\ &\quad + 2(-BX^2Z + GY^2 + HYZ + FXY)ZX \\ &\quad + 2(-CXY + HZ^2 + FZX + GZY)XY, \end{aligned}$$

and subtracting therefrom the function $(A, \dots \dots)$, which is

$$\begin{aligned} &= (BP + CY^2 + (AB - F^2)X^2 \\ &\quad + (CY^2 + AB - F^2)Y^2 \\ &\quad + (AY^2 + BX^2 - H^2)Z^2 \\ &\quad - 2(FY^2 + HZ + EXY)YZ \\ &\quad - 2(GZ^2 + EX + EXY)ZX \\ &\quad - 2(HX^2 + FEZ + EXY)XY, \end{aligned}$$

the difference is found to be

$$\begin{aligned} &= a(L \dots \dots YZ, ZP + GLY + GLZ) \\ &\quad + b(L \dots \dots EX, X^2 + EXY) \\ &\quad + c(L \dots \dots XY, Y^2 + EXY) \\ &\quad - 2f(LX + BLY + GLZ)YZ \\ &\quad - 2g(LX + BLY + GLZ + bLY)ZX \\ &\quad - 2h(LX + GLY + HLY)XY. \end{aligned}$$

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which, on account of $(A \dots \sim \xi X, Y \sim LZ)^2 = 0$, and $A + B + C = 0$, reduces itself to

$$(A \dots \sim \dots) V.$$

49. We have

$$\begin{aligned} Aa + Bb + Cg &= a(BX - 2g)V \\ &+ b(gLY - fY - a(bh)) \\ &+ p(VZ - bLX - cbZ)Y \\ &= X(gb - bgy) \\ &+ V(gy - py - (pl + g + h + hh) + L LX \\ &+ Z(bn - h - (bh + b)b + ff)b), \end{aligned}$$

or, observing that in the coefficients of Y and Z the second terms each vanish, this is

$$Aa + Bb + Cg = X(bp - g)h + Y(ga - h)b + Z(\xi b - ha);$$

and similarly

$$Ha + Bb + Fg = X(gh - f) + Y(h - bh - c) + Z(\xi h - gg),$$

$$Gg + Fb + Cg = X(gh - g) + Y(gb - h)b + Z(gf - gg).$$

Adding these equations, the coefficient of X is the difference of two expressions each of which vanishes; and the like as regards the coefficients of Y and Z ; that is, we have

$$(A \dots \sim \dots) = 0,$$

and consequently

$$2 \begin{vmatrix} LX, & LY, & LZ \\ X, & Y, & Z \\ LX, & LY, & LZ \end{vmatrix} = (A \dots \sim \dots) = -(A \dots \sim \xi X, LY, LZ)^2,$$

the required relation.

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On the Centro-Surface of an Ellipsoid.

[From the Transactions of the Cambridge Philosophical Society, vol. XII. Part i. (1873),
pp. 319–365. Read March 7, 1870.]

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THE Centro-surface of any given surface is the locus of the centres of curvature of the given surface, or say it is the locus of the intersections of consecutive normals, (the normals which intersect the normal at any particular point of the surface being those at the consecutive points along the two curves of curvature respectively which pass through the point on the surface). The terms, normal, centre of curvature, curve of curvature, may be understood in their ordinary sense, or in the generalised sense referring to the case where the Absolute (instead of being the imaginary circle at infinity) is any quadric surface whatever; viz. the normal at any point of a surface is here the line joining that point with the pole of the tangent plane in respect of the quadric surface called the Absolute: and of course the centre of curvature and curve of curvature refer to the normal as just defined.

The question of the centro-surface of a quadric surface has been considered in the two points of view, viz. 1^o, when the terms “normal,” &c. are used in the ordinary sense, and the equation of the quadric surface (assumed to be an ellipsoid) is taken to be

$$\frac{X^2}{a^2} + \frac{Y^2}{b^2} + \frac{Z^2}{c^2} = 1;$$

2^o, when the Absolute is the surface $X^2 + Y^2 + Z^2 + W^2 = 0$, and the equation of the quadric surface is taken to be $\alpha X^2 + \beta Y^2 + \gamma Z^2 + \delta W^2 = 0$: in the first of them by Salmon, *Quart. Math. Jour.*, t. II. pp. 217–222 (1858), and in the second by Clebsch, *Crelle*, t. LXII. pp. 64–107 (1863): see also Salmon’s *Solid Geometry*, 2nd Ed. 1865, pp. 143, 402, &c. In the present Memoir, as shown by the title, the quadric surface is taken to be an Ellipsoid; and the question is considered exclusively from the first point of view: the theory is further developed in various respects, and in particular as regards the nodal curve upon the centro-surface: the distinction of real and